EmPay – Smart Human Resource Management System

The Challenge

EmPay – Simplifying HR & Payroll Operations for Smarter Workplaces

Vision & Mission

EmPay aims to modernize and simplify how organizations manage people, processes, and payroll through a comprehensive, all-in-one **Human Resource Management System (HRMS)**. The platform is designed to provide a clean, reliable, and user-friendly experience for both employees and administrators—enabling seamless collaboration in managing attendance, leave, payroll, and analytics from a unified interface. By focusing on robust, scalable, and thoughtfully engineered solutions, EmPay strives to reduce manual dependency, improve transparency, and empower organizations—especially startups, institutions, and SMEs—to make informed, data-driven workforce decisions.

Problem Statement

Develop a working system, focusing on the following modules and flows:

User & Role Management

- User registration and login functionality.
- Role-based access (**Employee / HR Officer/ Admin / Payroll Officer**).
- Editable profile management.

Attendance & Leave Management

- Employees can mark attendance and view their daily/monthly logs.
- Leave application, approval, and rejection workflows.

Payroll Management

- Basic payroll module showing salary breakdown, deductions, and payout summary.
- Option for Admin/Payroll Officer to generate or edit monthly reports and payslips.

Dashboard & Analytics

- Display of attendance, leaves, and payroll metrics through charts or summaries.
- Admin overview of employee data and overall HR statistics.

Roles & Responsibilities

1. Admin

- Register on the portal and manage user accounts.
- Create, read, update, and delete data across all modules.
- Manage user roles in settings.
- Oversee all activities and ensure smooth system operations.
- No access limitations.

2. Employee

- Apply for time off and see time off status.
- View personal attendance and performance records.
- Can access the employee directory and view individual records but cannot modify them.
- Cannot access settings, payroll, reports, salary info.

3. HR Officer

- Create and update employee profiles and details.
- Monitor attendance records of all employees.
- Manage & Allocate new leaves to employees.
- Cannot access payroll data or access system settings.

4. Payroll Officer

- Approve or reject time-off requests.
- Generate payslip and leave reports.
- Cannot create or modify employee data or access system settings, except for managing salary-related information.
- Manage Payroll, Time Off and Reports.
- Can access attendance.

Terminologies

1. Payroll

The process of calculating and distributing employee salaries, wages, bonuses, and deductions. All payroll-related calculations are based on **employee attendance records** for the given pay period.

2. Payrun

A specific payroll cycle or period during which salaries are processed and paid to employees. The payrun amount is calculated based on attendance, approved leaves, and applicable deductions.

3. Payslip

An official document provided to an employee for each payrun, showing a detailed breakdown of earnings, deductions, and net pay. Payslips are automatically generated once payroll is processed.

4. Time-Off

A period during which an employee is officially permitted to be absent from work (such as vacation, sick leave, or personal leave). Approved time-offs are factored into payroll calculations.

5. Wage

The monetary compensation paid to an employee for their work during a specific period. Wages are calculated based on **attendance, working hours, and approved leaves**.

6. Provident Fund (PF) Contribution

A retirement benefit scheme in which both the **employee and employer** contribute a fixed percentage (commonly 12%) of the employee's basic salary every month. PF is calculated based on the **basic salary** and serves as long-term savings for the employee.

7. Professional Tax

A small monthly tax levied by the state government on individuals earning income through employment or profession. It is deducted directly from the **gross salary** as part of monthly payroll processing.

Deliverables

- Source code hosted on Git repository (with meaningful commits)

Why is this Hackathon Problem Important?

- Students will learn real-world **ERP workflows**.
- Understand how modules talk to each other for **Eg** (Employees → Attendance → Payroll).
- Practice problem-solving using business logic, not just coding.

Mockup:- https://link.excalidraw.com/l/65VNwvy7c4X/7gxoB8JymIS